



DURBAN UNIVERSITY OF TECHNOLOGY

Improving Cloud file sharing with two – factor authentication for freelances

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DECLARATION

I, Bandile Cele, declare that:

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1. Research Context

One of the major security concerns in cloud file sharing is weak authentication mechanisms. A lot of freelancers rely on password based security, which can be easily compromised with attacks such as phishing attacks, brute-force attacks and credentials leaks. The lack of security layers of protection grant attackers access to cloud-stored files, exposing private and sensitive information.

To minimise these risks, Two-Factor Authentication (2FA) is a highly effective security measure. Unlike traditional password-based access, 2FA requires users to verify their identity through an additional step, such as a **one-time password (OTP), biometric verification, or authentication apps**. This extra layer of security significantly reduces the chances of **unauthorized access**, ensuring that only **verified users** can retrieve or share files.

For freelancers who often work remotely and share files with multiple clients, **implementing 2FA in cloud file sharing is essential**. It enhances **data security, protects client trust, and reduces the risk of cyberattacks**. By integrating 2FA into cloud storage systems, freelancers can **safeguard their data and maintain the confidentiality and integrity** of their shared files.

1.1 Research scope

This research will mainly focuses on securing cloud file sharing by the integration of Two-Factor Authentication(2FA).This study is only limited to freelancers who use cloud storage solutions (such as Google Drive, Dropbox, and OneDrive) to store and share sensitive files. The proposed solution will be designed to enhance the security of cloud file sharing by requiring additional authentication steps before granting access.

1.2 Research Details

Keywords : “Cloud-file sharing security”, “Two-factor-authentication”

Database: Ieee Xplore

Result : 1,667, 1,990

Links : [IEEE Xplore Search Results](#)[IEEE, Xplore Search Results](#), [Google Scholar](#)

2. Literature review

2.1 State of the art

Cloud file-sharing security is a growing concern due to risks like unauthorized access and data breaches. While existing studies have explored **Two-Factor Authentication (2FA)** for cloud security, many solutions, such as certificate-based authentication and proxy re-encryption, are complex or require high computational power, making them unsuitable for freelancers. Most research focuses on corporate environments, leaving freelancers with limited security options. To address this gap, this study proposes a **freelancer-friendly 2FA model** that prioritizes **ease of use, lightweight encryption, and seamless cloud integration**. The model will be tested for **security strength, authentication speed, and user experience** to ensure a practical and effective solution.

Author	Problem	Database/API	method	Findings	Limitations
1.Dr.K. Subramanian al et, 2017	Security risks with data sharing in cloud storage with potential for unauthorized access by malicious insiders.	-YouTube database for video files -Statistical Analysis system(SAS) for commercial bank	-Index-based cryptographic data slicing -Encryption -File Retrieval protocol	-Improved security -Efficiency in data handling -Future improvements	-Vulnerability to certain attacks -User errors and insufficient automation
Solonas Goustoris et al, 2023	Current system often don't keep sensitive data private because they use public block-chain and information on block-chain is publicly visible	N/A	-Block-chain structure -RSA encryption	- Confidentiality and integrity -Trust and search when using encryption -scalability	-Complexity in setting it up -Performance issue with large files -potential vulnerability in Blockchain
V. Rajkumar, M. Prakash, V. Vennila, 2022	Unauthorised user might access private files -Lack of strong authentication	-Code DxAPI	-GQ protocol -RSA algorithm -Access List(ACL)	-Improved security using GQ protocol -strong protection data integrity	--Complexity in implementation

	makes it easy for hackers to break in		-STPN simulation		
Parashiva Murthy Basavanapara Maddmadappa, et al, 2022	Lack of security through file sharing across multi cloud storage services	N/A	Dynamic File slicing -Segregation of Data storage	Effective dynamic file sharing -Performance improvement -Resistance to malicious File attacks	-Complexity of implementation -Need for continuous monitory
Sandesh Anchor,Hrishiva Patel, Sansal Hussain, 2022	The complexity and risks associated with data breaches and Inadequate security strategy	N/A	-Data Encryption -Data Classification	-Increase in cyber threats -identify management to ensure only authorised users have access	-Complexity of cloud environment -Insiders threats -Inadequate training and awareness
Alexandra Dmitrienko et al, 2014	The two factor authentication schemes are vulnerable to attack that involve the user interacting with Trojan horse malware and weak OTP management	-VirusTotal -Contagioudump project	-Behavior monitoring malwares that may be able to attack -Reverse engineering malware -Grouping similar malware using hierachiry	Weak OTP management 2FA deactivation and recovery weaknesses make the system more vulnerable to attacks	Assume pre-existing malware infection with are well known and does not
Ammar Odeh,Quson Abu Al-Haijah et al, 2024	Traditional cloud file sharing methods have weak encryption or improper by management which can expose confidential data.	AWS identity and Access management(IAM) -Azure sentinel API	-Access control and identity management		
V. Rajkumar,M. Prakash, V.Vennila,2022	- Unauthorized user might access private files -Lack of strong authentication makes it easy for hackers to break in	N/A	-Dynamic File slicing -Segregation of data storage	-Effective dynamic file sharing -Performance improvement -Resistance to malicious file attacks	-Complexity of implementation -Need for continuous monitoring

Abhishok iche, Abhishok Mhamane, Ayan Thaikh,2022	Methods of file sharing lack user control and awareness regarding the type of files being shared, potentially leading to security risk	N/A	AES Encryption -key management techniques -User authentication and access control -Key management techniques	-Effectiveness of AES encryption was found to be an efficient method for security file at rest within cloud storage -The standard cloud-based files sharing system without security measures are vulnerable to various threats.	-Performance overhead -Complexity of key management
Shruti Kanatt, Amey Jadhav, Prachi Talwar, 2022	-Security risk and data leaks possibilities associated with cloud storage	N/A	ECC(Elliptic Curve Cryptography -Separation of functions using different servers -Hybrid cryptography approach(RSA and AES)	-The importance of robust user authentication methods alongside encryption to secure data. -Separation of server functions. -Cloud-storage is more cost-effective than traditional physical storage.	-Complexity of process -Connectivity and synchronization issues.

2.2 Encryption Techniques for Cloud Security

Encryption is a fundamental method used to protect cloud-stored files from unauthorized access. Various encryption techniques, including RSA, AES, and ECC, have been implemented to enhance data confidentiality and integrity. Some researchers have also explored hybrid cryptography, combining multiple encryption methods for stronger security. Rajkumar et al. (2022) integrated RSA encryption with Access Control Lists (ACLs) to ensure that only authorized users could decrypt stored files. Similarly, Kanatt et al. (2022) used a hybrid approach (RSA + AES) to balance security and efficiency.

However, encryption-based solutions often come with challenges .A major limitation of encryption methods is key management complexity, which can make implementation difficult

for non-expert users. Additionally, encryption introduces performance overhead, especially for large files, which can slow down file-sharing and retrieval processes (Iche et al., 2022).

2.3 Authentication and Access Control Mechanisms

Authentication and access control play a crucial role in securing cloud storage by ensuring that only authorized users can access sensitive data. Various methods, such as Two-Factor Authentication (2FA), Guillou-Quisquater (GQ) protocol, and Identity and Access Management (IAM), have been explored. Rajkumar et al. (2022) implemented GQ protocol with RSA encryption, which enhanced authentication security through multi-step verification. Similarly, Odeh et al. (2024) utilized AWS IAM and Azure Sentinel API for strict identity management and access control in cloud environments.

Despite these advancements, authentication methods are not without limitations. "Some 2FA schemes are vulnerable to Trojan horse malware and weak OTP management, which attackers can exploit" (Dmitrienko et al., 2014). Additionally, complex authentication systems can reduce user convenience, leading to low adoption rates among freelancers and small businesses.

2.4 Secure Cloud File Sharing Strategies

Cloud file-sharing requires strong security measures to prevent data leaks and unauthorized modifications. Several researchers have explored dynamic file slicing, data segregation, and blockchain-based security to enhance file-sharing security. Basavanapara et al. (2022) introduced dynamic file slicing and data segregation, which improved file-sharing performance and resistance to malicious attacks. Goustoris et al. (2023) leveraged blockchain encryption to ensure trust and confidentiality in file sharing.

However, these techniques also face challenges. While dynamic file slicing improves security, it increases implementation complexity and requires continuous monitoring (Basavanapara et al., 2022). Similarly, blockchain-based solutions suffer from performance issues, especially when handling large file sizes, and can be complex to set up (Goustoris et al., 2023).

2.5 Synthesis

Research highlights unauthorized access as a major cloud security risk, with multi-factor authentication and encryption being key protective measures (Rajkumar et al., 2022; Kanatt et al., 2022; Abhishok et al., 2022). AES, RSA, and ECC encryption methods are widely used, with some studies proposing blockchain-based security for enhanced data confidentiality (Goustoris et al., 2023).

File segmentation and dynamic data slicing further improve security by distributing data across multiple locations, reducing the risk of full file exposure in case of a breach (Maddmadappa et al., 2022; Rajkumar et al., 2022). Different studies focus on blockchain security (Goustoris et al., 2023), 2FA vulnerabilities (Dmitrienko et al., 2014), and cloud-specific security tools (Odeh & Al-Haijah, 2024), highlighting varied approaches to cloud security.

While encryption, authentication, and access control remain core security measures, a multi-layered strategy integrating blockchain, dynamic file slicing, hybrid encryption, and improved authentication methods is essential to address evolving cloud security threats.

2.6 Research Gap

Despite advancements in cloud security, freelancers and small businesses continue to face major challenges. Encryption techniques such as RSA and AES provide strong protection, but they are often complex, resource-intensive, and difficult to implement (Rajkumar et al., 2022).

Two-Factor Authentication (2FA) enhances security, but many existing implementations suffer from OTP vulnerabilities, malware risks, and difficult setups, leading freelancers to avoid them. Secure file-sharing methods like dynamic file slicing and blockchain encryption offer better security, but they require technical expertise, high computational power, and lack scalability (Basavanapara et al., 2022; Goustoris et al., 2023).

The biggest challenge is developing a security solution that maintains strong protection while being simple and accessible—something freelancers can easily adopt without compromising their productivity.

3. Research problem

Many freelancers rely on password-based authentication, which is prone to security breaches. While 2FA offers better protection, its complexity and lack of seamless integration with cloud platforms discourage adoption. As a result, files remain at risk of unauthorized access and cyber threats. This research aims to fill this gap by creating an authentication system that ensures security while maintaining ease of use.

4. The Aim

This study aims to design and develop a user-friendly and efficient Two-Factor Authentication (2FA) system that strengthens cloud file-sharing security for freelancers while minimizing complexity and improving adoption.

4.1 The Objectives

- ☐ To review the limitations of current Two-Factor Authentication (2FA) methods in cloud file sharing and understand the challenges freelancers face in using them.
- ☐ To design a more user-friendly and efficient 2FA system that enhances security without adding unnecessary complexity.
- ☐ To develop a functional prototype of the improved 2FA system tailored for freelancers.

□ To evaluate the effectiveness of the new 2FA system in terms of security, usability, and seamless integration with cloud storage platforms.

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